

Mathematical Methods I

Differential 1-forms

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1. Covectors and 1-forms

A tangent vector describes a direction and a magnitude at a point. A 1-form evaluates that vector and produces a real number in a linear way.

Definition 1.1 (Covector and covector field)

Let $\Omega \subseteq \mathbb{R}^n$ be open. A **covector** at $p \in \Omega$ is a linear map

$$\omega_p : T_p\Omega \longrightarrow \mathbb{R}.$$

The set of these functionals is the **cotangent space** $T_p^*\Omega = (T_p\Omega)^*$, the dual of the tangent space. A **differential 1-form** ω assigns a covector ω_p to each point p , with smooth coefficient functions in the case of a smooth form.

Linearity refers to vectors based at the same point: for $u_p, v_p \in T_p\Omega$ and $a, b \in \mathbb{R}$,

$$\omega_p(au_p + bv_p) = a\omega_p(u_p) + b\omega_p(v_p).$$

Operations with 1-forms

For 1-forms ϕ, ψ and a scalar function f , define

$$\begin{aligned} (\phi + \psi)_p(v_p) &= \phi_p(v_p) + \psi_p(v_p), \\ (f\phi)_p(v_p) &= f(p)\phi_p(v_p). \end{aligned}$$

The factor f is evaluated at the base point of the vector; it is not differentiated.

Evaluation on a vector field

If X is a vector field, $\omega(X)$ is the scalar function

$$(\omega(X))(p) = \omega_p(X(p)).$$

In particular, $\omega_p(v_p)$ is a number, whereas $\omega(X)$ is a function of p .

Linearity over functions

For vector fields X, Y and scalar functions f, g ,

$$\omega(fX + gY) = f\omega(X) + g\omega(Y).$$

Thus, a 1-form transforms vector fields into functions by pointwise evaluation.

2. Coordinate differentials and the dual basis

Let $x^1, \dots, x^n$ be the Cartesian coordinate functions. We write

$$U_j = \partial_j = \frac{\partial}{\partial x^j}, \quad v_p = \sum_{j=1}^n v^j \partial_j|_p.$$

The superscripts in x^i and v^i are indices, not powers.

Definition 2.1 (Differential of a function)

For a differentiable function $f : \Omega \rightarrow \mathbb{R}$, its differential at p is

$$(df)_p(v_p) = v_p[f].$$

The differentiability of f guarantees that this expression is linear in v_p . If f is smooth, df is a smooth 1-form. Applied to $f = x^i$,

$$(dx^i)_p(v_p) = v_p[x^i] = v^i.$$

The coordinate 1-form dx^i selects the i th component of the vector.

Proposition 2.2 (Duality of the bases)

At each point, $\{(dx^1)_p, \dots, (dx^n)_p\}$ is the dual basis of $\{\partial_1|_p, \dots, \partial_n|_p\}$. Indeed,

$$(dx^i)_p(\partial_j|_p) = \partial_j|_p[x^i] = \frac{\partial x^i}{\partial x^j}(p) = \delta_j^i, \quad \delta_j^i = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

As an identity between functions, we write $dx^i(U_j) = \delta_j^i$.

In Cartesian coordinates, the value $dx^i(v_p) = v^i$ does not explicitly depend on p when the vector components are fixed.

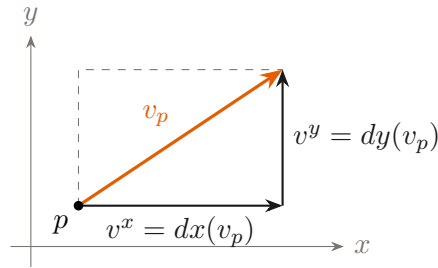


Figure 1: The forms dx and dy return the components of the vector based at p . The horizontal and vertical arrows represent vector components, not the 1-forms themselves.

3. Expression and evaluation of a 1-form

Proposition 3.1 (Coordinate expression)

Every 1-form can be uniquely expressed as

$$\omega = \sum_{i=1}^n a_i dx^i,$$

where the a_i are scalar functions. Its coefficients are recovered by evaluating it on the tangent basis:

$$a_i = \omega(U_i).$$

Proof and evaluation formula

At a point p , let $v_p = \sum_j v^j \partial_j|_p$. By linearity and duality,

$$\begin{aligned} \omega_p(v_p) &= \left(\sum_i a_i(p) (dx^i)_p \right) \left(\sum_j v^j \partial_j|_p \right) \\ &= \sum_i \sum_j a_i(p) v^j (dx^i)_p (\partial_j|_p) \\ &= \sum_i \sum_j a_i(p) v^j \delta_j^i = \sum_i a_i(p) v^i. \end{aligned}$$

In particular,

$$\omega(U_j) = \sum_i a_i dx^i(U_j) = \sum_i a_i \delta_j^i = a_j.$$

Conversely, defining $a_i = \omega(U_i)$, the form $\sum_i a_i dx^i$ agrees with ω on every basis vector and, by linearity, on every vector. This proves the existence and uniqueness of the expression.

Evaluation on a vector field

If $X = \sum_i X^i \partial_i$, then

$$\omega(X) = \sum_i a_i X^i.$$

Both sides are functions; evaluation at p gives $\omega_p(X(p))$.

Example 3.2 (A function and its value at a point)

In $\mathbb{R}^2$, let $\omega = y dx + 2x dy$ and $X = x \partial_x - \partial_y$. Then

$$\omega(X) = xy - 2x.$$

For $p = (1, 3)$, $\omega_p = 3(dx)_p + 2(dy)_p$ and $X(p) = (1, -1)_p$, so $\omega_p(X(p)) = 3 - 2 = 1$. Moreover, $\omega(\partial_x) = y$ and $\omega(\partial_y) = 2x$ recover the coefficients.

4. The differential df and exact forms

Theorem 4.1 (Coordinate expression of the differential)

For a differentiable function $f : \Omega \rightarrow \mathbb{R}$,

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i.$$

A 1-form that can be written as df throughout its domain is called **exact**. The function f is a potential for that form.

Proof by evaluation

For any $v_p = \sum_j v^j \partial_j|_p$, the definition of the differential gives

$$(df)_p(v_p) = v_p[f] = \sum_i v^i \frac{\partial f}{\partial x^i}(p).$$

On the other hand, using duality,

$$\begin{aligned} \left(\sum_i \frac{\partial f}{\partial x^i}(p) (dx^i)_p \right) (v_p) &= \sum_i \sum_j \frac{\partial f}{\partial x^i}(p) v^j \delta_j^i \\ &= \sum_i \frac{\partial f}{\partial x^i}(p) v^i. \end{aligned}$$

The two forms agree on all tangent vectors and are therefore equal. Equivalently, their coefficients are obtained as

$$df(U_i) = U_i[f] = \frac{\partial f}{\partial x^i}.$$

Example 4.2 (Computing df)

If $f(x, y, z) = x^2y + z^2$, then

$$df = 2xy \, dx + x^2 \, dy + 2z \, dz.$$

For $p = (1, 2, -1)$ and $v_p = (2, -1, 3)_p$,

$$(df)_p(v_p) = 4(2) + 1(-1) - 2(3) = 1.$$

This gives the directional derivative of f along the vector v_p .

Differential, vector field, and curve

For any vector field X , $df(X) = X[f]$. If α is a differentiable curve contained in Ω , the chain rule states that

$$(df)_{\alpha(t)}(\alpha'(t)) = \frac{d}{dt}(f(\alpha(t))).$$

Evaluating df on the velocity measures how f changes along the curve.

5. The operator d : products and compositions

The operator d transforms a scalar function f into the 1-form df . The usual differentiation rules thus become equalities between 1-forms. We assume that the functions are differentiable on their domains.

Proposition 5.1 (Linearity and the product rule)

For $f, g : \Omega \rightarrow \mathbb{R}$ and constants $a, b \in \mathbb{R}$,

$$d(af + bg) = a df + b dg, \quad d(c) = 0 \quad (c \text{ constant}).$$

The **Leibniz rule** states that

$$d(fg) = f dg + g df.$$

The factors f and g multiply 1-forms. At a point p , this equality means

$$(d(fg))_p = f(p)(dg)_p + g(p)(df)_p.$$

Why the product rule holds

For any tangent vector v_p , we apply the definition of the differential and the product rule for directional derivatives:

$$\begin{aligned} (d(fg))_p(v_p) &= v_p[fg] \\ &= f(p) v_p[g] + g(p) v_p[f] \\ &= f(p)(dg)_p(v_p) + g(p)(df)_p(v_p). \end{aligned}$$

Since both 1-forms give the same result on every v_p , they are equal. Linearity is justified in the same way.

Example 5.2 (Product of two functions)

Let $f(x, y) = x^2$ and $g(x, y) = y$. First, we compute

$$df = 2x dx, \quad dg = dy.$$

Then we apply the rule:

$$\begin{aligned} d(x^2 y) &= x^2 d(y) + y d(x^2) \\ &= x^2 dy + 2xy dx. \end{aligned}$$

This gives the term corresponding to $x^2 y$ in Example 4.2. Since $d(z^2) = 2z dz$, linearity gives

$$d(x^2 y + z^2) = 2xy dx + x^2 dy + 2z dz.$$

In $f dg$, the function multiplies the form: $(f dg)_p(v_p) = f(p)(dg)_p(v_p)$. Two differentials are not multiplied together; $d(fg)$ is a 1-form.

5. The operator d (continued)

Proposition 5.3 (Chain rule)

Let $f : \Omega \rightarrow \mathbb{R}$ and $h : A \rightarrow \mathbb{R}$, where $A \subseteq \mathbb{R}$ is open and $f(\Omega) \subseteq A$. Then

$$d(h \circ f) = (h' \circ f) df = h'(f) df.$$

The outer function h is differentiated, its derivative is evaluated at f , and the result is multiplied by the differential of the inner function.

Proof by evaluation

For every tangent vector v_p , the ordinary chain rule gives

$$(d(h \circ f))_p(v_p) = v_p[h \circ f] = h'(f(p)) v_p[f] = h'(f(p)) (df)_p(v_p).$$

This proves the equality of the 1-forms.

Example 5.4 (A composition step by step)

For $F(x, y) = \sin(x^2 y)$, the inner function is $f(x, y) = x^2 y$ and the outer function is $h(u) = \sin u$. We have

$$h'(u) = \cos u, \quad df = 2xy dx + x^2 dy.$$

Therefore,

$$dF = \cos(x^2 y)(2xy dx + x^2 dy).$$

The factor $\cos(x^2 y)$ multiplies the entire 1-form df .

Composition with several inner functions

If $F = H(f_1, \dots, f_m)$, where H is differentiable on an open set containing the image of $(f_1, \dots, f_m)$, then

$$dF = \sum_{a=1}^m \frac{\partial H}{\partial u^a}(f_1, \dots, f_m) df_a.$$

For example, for $F = e^{xy} + z^2$, take $H(u, v) = e^u + v^2$, $f_1 = xy$, and $f_2 = z$. Thus,

$$dF = e^{xy} d(xy) + 2z dz = e^{xy}(y dx + x dy) + 2z dz.$$

Useful consequences

$$d(f^k) = k f^{k-1} df \quad (k = 1, 2, \dots), \quad d(e^f) = e^f df.$$

Combining the product and composition rules, where $g \neq 0$,

$$d\left(\frac{f}{g}\right) = \frac{1}{g} df - \frac{f}{g^2} dg = \frac{g df - f dg}{g^2}.$$

6. Indices and coordinate changes

The components of a vector and a 1-form change with the coordinates, although the geometric object and its evaluation remain the same.

Einstein summation convention

An index repeated once as a superscript and once as a subscript indicates summation:

$$X = X^i \partial_i, \quad \omega = \omega_i dx^i, \quad \omega(X) = \omega_i X^i.$$

The repeated index is a **dummy index**: it may be renamed without changing the expression. Free indices must match on both sides of an equality.

The vector components X^i are **contravariant**; the components ω_i of a 1-form are **covariant**. The tangent basis ∂_i and the cotangent basis dx^i transform so as to compensate for the change in their respective components.

A fixed vector, different coordinate descriptions

Think of a tangent vector as a fixed arrow at a point p , for example the velocity of a particle at one instant. The arrow is the geometric object; its components depend on the coordinates used to describe it. Let $\alpha(t)$ be a curve with $\alpha(0) = p$ and $v_p = \alpha'(0)$. In coordinates x and u , the same motion has components

$$v_x^i = \left. \frac{d(x^i \circ \alpha)}{dt} \right|_{t=0}, \quad v_u^a = \left. \frac{d(u^a \circ \alpha)}{dt} \right|_{t=0}.$$

Applying the chain rule to $u^a = u^a(x)$ gives

$$v_u^a = \frac{\partial u^a}{\partial x^i}(p) v_x^i.$$

Thus, the transformation law for vector components is simply the chain rule for the velocity of the same curve. With the convention $J = \partial x / \partial u$ used below, this is $v_u = J(p)^{-1} v_x$. The Jacobian converts the components of a tangent vector locally at p .

6. Indices and coordinate changes (continued)

General rules without matrix ambiguity

Let $x = (x^1, \dots, x^n)$ and $u = (u^1, \dots, u^n)$ be two local coordinate systems, with invertible change of coordinates $x = x(u)$. Set

$$J^i_a = \frac{\partial x^i}{\partial u^a}, \quad (J^{-1})^a_i = \frac{\partial u^a}{\partial x^i}.$$

Transformation of bases and components

By the chain rule,

$$\begin{aligned} \partial_{u^a} &= \frac{\partial x^i}{\partial u^a} \partial_{x^i}, & du^a &= \frac{\partial u^a}{\partial x^i} dx^i, \\ X_u^a &= \frac{\partial u^a}{\partial x^i} X_x^i, & \omega_{u,a} &= \frac{\partial x^i}{\partial u^a} \omega_{x,i}. \end{aligned}$$

Writing each list as a column, these identities become

$$\partial_u = J^T \partial_x, \quad du = J^{-1} dx, \quad X_u = J^{-1} X_x, \quad \omega_u = J^T \omega_x.$$

Invariance of evaluation

Contracting a covariant index with a contravariant one gives

$$\omega_{u,a} X_u^a = \frac{\partial x^i}{\partial u^a} \omega_{x,i} \frac{\partial u^a}{\partial x^j} X_x^j = \delta_j^i \omega_{x,i} X_x^j = \omega_{x,i} X_x^i.$$

An index repeated twice as a superscript or twice as a subscript does not represent a contraction under this convention. For example, the inner product is written $g_{ij} A^i B^j$; in an orthonormal Cartesian basis it can be simplified to $\sum_i A^i B^i$, because there $g_{ij} = \delta_{ij}$.

6. Connection with classical mechanics

Generalized coordinates and generalized velocities

In classical mechanics, a configuration is a point of configuration space. Generalized coordinates $q^1, \dots, q^m$ are local coordinates on that space; they may be lengths, angles, or other independent parameters. A motion is a curve $q(t)$, and its tangent vector is

$$\dot{\alpha}(t) = \dot{q}^a(t) \partial_{q^a}|_{\alpha(t)}, \quad dq^a(\dot{\alpha}) = \dot{q}^a.$$

The numbers $\dot{q}^a$ are the **generalized velocities**. They are coordinate components, so they need not be speeds measured along unit vectors. For example, $\dot{\theta}$ is an angular velocity, whereas $r\dot{\theta}$ is a tangential speed. For a particle with a time-independent position map $\mathbf{r} = \mathbf{r}(q)$, the physical velocity is

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \frac{\partial \mathbf{r}}{\partial q^a} \dot{q}^a.$$

For an unconstrained particle with $m = n$ and an invertible coordinate map, this is precisely $\mathbf{v}_x = J\dot{\mathbf{q}}$, with $J = \partial x / \partial q$, or $\dot{\mathbf{q}} = J^{-1}\mathbf{v}_x$. With constraints, m may be smaller than the number of ambient Cartesian coordinates: the chain rule still holds, but the position Jacobian is then rectangular and has no ordinary inverse. For explicitly time-dependent maps $\mathbf{r}(q, t)$, the velocity also includes $\partial \mathbf{r} / \partial t$.

Forces as 1-forms: virtual work and power

For a force $\mathbf{F}$ on a particle, the Euclidean inner product defines a work 1-form on configuration space:

$$\eta = \mathbf{F} \cdot d\mathbf{r} = Q_a dq^a, \quad Q_a = \mathbf{F} \cdot \frac{\partial \mathbf{r}}{\partial q^a}.$$

7. Example: polar coordinates

Consider $x = r \cos \theta$, $y = r \sin \theta$, with $r > 0$ and a smooth local choice of the angle. The origin lies outside this chart.

Jacobian and inverse matrix

With $J = \partial(x, y)/\partial(r, \theta)$,

$$J = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}, \quad J^{-1} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta/r & \cos \theta/r \end{pmatrix}.$$

We have $\det J = r \neq 0$.

Tangent basis and vector components

$$\begin{aligned} \partial_r &= \cos \theta \partial_x + \sin \theta \partial_y, \\ \partial_\theta &= -r \sin \theta \partial_x + r \cos \theta \partial_y, \\ V^r &= \cos \theta V^x + \sin \theta V^y, \\ V^\theta &= -\frac{\sin \theta}{r} V^x + \frac{\cos \theta}{r} V^y. \end{aligned}$$

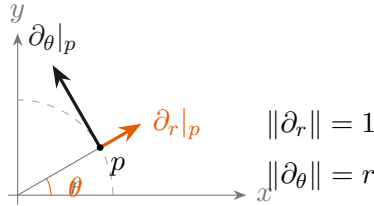


Figure 2: Polar coordinate basis, illustrated at $r = 2$. The vector ∂_θ is tangent to the circle and has length r ; it is not generally a unit vector.

Cotangent basis and 1-form components

$$\begin{aligned} dr &= \cos \theta dx + \sin \theta dy, \\ d\theta &= -\frac{\sin \theta}{r} dx + \frac{\cos \theta}{r} dy, \\ \omega_r &= \cos \theta \omega_x + \sin \theta \omega_y, \\ \omega_\theta &= -r \sin \theta \omega_x + r \cos \theta \omega_y. \end{aligned}$$

These formulas verify that $dr(\partial_r) = d\theta(\partial_\theta) = 1$ and $dr(\partial_\theta) = d\theta(\partial_r) = 0$. With the columns and Jacobian defined above, the correct relation is

$$\begin{pmatrix} dr \\ d\theta \end{pmatrix} = J^{-1} \begin{pmatrix} dx \\ dy \end{pmatrix}.$$

8. Verification in both coordinate systems

Example 8.1 (A 1-form and a vector)

Consider

$$\omega = x dx + y dy, \quad V = \partial_x.$$

In Cartesian coordinates, $\omega(V) = x$. In polar coordinates, the components of the form are

$$\begin{aligned}\omega_r &= x \cos \theta + y \sin \theta = r, \\ \omega_\theta &= -rx \sin \theta + ry \cos \theta = 0.\end{aligned}$$

Therefore, $\omega = r dr$. The components of the same vector are

$$V^r = \cos \theta, \quad V^\theta = -\frac{\sin \theta}{r}.$$

Evaluation in the polar basis gives

$$\omega(V) = \omega_r V^r + \omega_\theta V^\theta = r \cos \theta = x.$$

The results agree: evaluation does not depend on the coordinates.

The same form as an exact differential

Define

$$f(x, y) = \frac{x^2 + y^2}{2}.$$

In Cartesian coordinates, $df = x dx + y dy$. In polar coordinates, the same function is expressed as $f = r^2/2$, so

$$df = r dr.$$

The two expressions describe the same 1-form. The potential f exists on all of $\mathbb{R}^2$, although polar coordinates are used only locally away from the origin.

Differential and gradient

The differential df is a covector. In Euclidean space, the inner product relates it to the gradient vector:

$$(df)_p(v_p) = \nabla f(p) \cdot v.$$

This identification uses the metric; the definition of df as a linear functional does not require it.